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# EduGrid Central

*A React-Powered E-Learning Platform*

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Project #70

 [edu-grid-co.vercel.app](https://edu-grid-co.vercel.app)

# Agenda

01

## Problem Statement

Why we need EduGrid Central

02

## Technology Stack

React 19 · Vite · React Router v7

03

## Key Features

8 must-have capabilities built

04

## Component Architecture

Modular, scalable design

05

## UI Walkthrough

Landing → Dashboard → Quiz → Certs

06

## Conclusion & Links

GitHub · Live Demo · References

# Problem Statement

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*The online education boom revealed serious UX gaps in existing platforms:*

## **Slow Navigation**

Multi-page reloads disrupt the learning flow

## **Poor Mobile UX**

Inconsistent UI across devices, poor responsiveness

## **No Progress Tracking**

Lack of transparent milestone visibility

## **Cluttered Catalogs**

Overwhelming course lists with no smart discovery

## **No Certificates**

No system to validate or reward learning outcomes

# Technology Stack

*Modern, production-grade front-end toolchain*



**React 19**

Component-based SPA



**Vite**

Lightning-fast build + HMR



**React Router v7**

Client-side routing



**Pure CSS**

Grid & Flexbox layouts



**Vercel**

CI/CD auto-deploy



**ESLint**

Code quality

# 8 Must-Have Features

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a

## Expandable Lesson List

Tiered modules — click to reveal video lectures & reading material

b

## Video Progress Tracker

Saves exact spot; undo accidental skips instantly

c

## Guided Quiz Creator

Step-by-step MCQ form with answer saving & layout management

d

## Lag-Free Gradebook

Scrollable table of all grades & milestones without slowdown

e

## Grade Sorter

Rank & filter scores by course category

f

## Student Access Hub

Central system verifying enrollment & content permissions

g

## Video Error Safety

Auto-displays text fallback if video fails to load

h

## Subtitle Customizer

Adjust subtitle size & style for any screen

# Component Architecture

## 📁 Directory Structure

```
EduGrid-Central/  
├── src/  
│   ├── components/  
│   ├── pages/  
│   ├── assets/  
│   └── App.jsx  
├── public/  
├── index.html  
└── vite.config.js
```

| Component             | Responsibility               | Type    |
|-----------------------|------------------------------|---------|
| <b>App.jsx</b>        | Root + Routes                | Root    |
| <b>Navbar</b>         | Top navigation links         | Layout  |
| <b>Hero</b>           | Landing CTA section          | Page    |
| <b>CourseGrid</b>     | Grid of courses with filters | Feature |
| <b>VideoPlayer</b>    | Video with progress tracking | Feature |
| <b>QuizPanel</b>      | MCQ quiz handler             | Feature |
| <b>GradebookPanel</b> | Scrollable grades table      | Feature |
| <b>Certificate</b>    | Certificate on completion    | Feature |

# UI Walkthrough

10 Screens · 1 Seamless SPA Experience



## Landing Page

Hero · Features · Pricing



## Login / Pricing

Auth form · Plan selector



## Dashboard

Progress cards · sidebar



## Courses Page

Grid catalog · filters



## Video Player

Tracker · fallback · subs



## Quiz Page

Guided MCQ creator



## Quiz Player

Step-by-step experience



## Student Data

Enrollment verification



## Certificates

Certificate gallery



## Certificate View

Styled cert + name

# Project Achievements

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8

Must-Have Features  
Implemented

10+

UI Screens  
Built

100%

Responsive  
Design

0

UI Framework  
Dependencies

## Engineering Principles Applied

- Component-Based Architecture — separation of concerns
- DRY Code — reusable components across the platform
- Performance-First — Vite HMR, client-side routing, lazy rendering
- CI/CD via Vercel — every push deploys automatically to production

# Conclusion

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- Fast, responsive SPA — React 19 + Vite delivers near-instant navigation
- Intuitive grid-based course discovery across all screen sizes
- Seamless client-side routing via React Router v7
- Complete learning lifecycle — progress tracking + certificate generation
- Clean, maintainable codebase with modular component architecture
- Auto CI/CD — GitHub push → Vercel production in seconds

## 🔗 Project Links

GitHub: [github.com/Yashhh710/EduGrid-Central](https://github.com/Yashhh710/EduGrid-Central)

Live Demo: [edu-grid-co.vercel.app](https://edu-grid-co.vercel.app)

## 📖 Tech References

[react.dev](https://react.dev) · [reactrouter.com](https://reactrouter.com) · [vitejs.dev](https://vitejs.dev)

[vercel.com/docs](https://vercel.com/docs) · [eslint.org](https://eslint.org) · [MDN Web Docs](https://mdn.com)



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# Thank You

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